

Portuguese post-OCR Resources for Text Optimisation

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Abstract

Optical Character Recognition (OCR) systems are designed to extract text from images. While typically optimised for modern documents, they often struggle when applied to historical documents due to older fonts, complex layouts, and physical degradation, which can result in noisy outputs. To reduce OCR errors, post-OCR algorithms are commonly used, however, their development and evaluation require image-transcription pairs. Compared to other European languages, there is a lack of transcribed documents for historical Portuguese, especially for texts predating the 19th century. To address this gap, we introduce Portuguese post-OCR Resources for Text Optimisation (PORTO), a dataset that spans from the 17th to the 20th centuries. PORTO contains 3,782 image-transcription pairs, along with OCR outputs from four different systems, providing a valuable resource for the development and evaluation of OCR and post-OCR methods tailored to historical Portuguese.

CCS Concepts

• Computing methodologies → Language resources; • Applied computing → Optical character recognition; Document scanning.

Keywords

Historical Portuguese Corpora, Post-OCR Correction resources, OCR evaluation

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1 Introduction

Historical documents are past documents about persons, places, events, laws, or other subjects, handwritten or mechanically written [40], that allow the past to be reconstructed and understood. These documents are typically stored in archives or libraries.

In the early 2000s, archives and libraries from many countries started digitising historical documents to improve their preservation [41]. Digitisation consists of converting a physical record to a machine-readable form, typically by scanning or photographing it. In some cases, algorithms such as Optical Character Recognition (OCR) are applied to extract text, further improving accessibility and

usability [29, 36, 40]. Digitising historical documents into machine-readable formats ensures their accessibility regardless of fragility, preventing physical damage while preserving their content even if the original is lost [40]. Moreover, it enhances research practices by enabling virtual archives to be hyperlinked to research projects, introducing a new form of citation that allows readers to access historical evidence directly [37]. When text is extracted, it also enables data mining and information retrieval techniques, allowing historians, archivists, and researchers to analyse vast collections of documents efficiently.

Unfortunately, OCR systems often struggle to extract text from historical documents due to physical deterioration, complex layouts, and limitations of OCR technology [13, 42, 43]. This is due to commercial OCR systems and open-source alternatives being primarily optimised for contemporary and general end-user documents [7]. As a result, when applied to historical texts, the output often contains significant noise, making search tools ineffective, reducing readability, and degrading the performance in downstream tasks [5, 12, 15–17, 23, 25, 26, 28, 51]. To address this issue, post-OCR processing systems are commonly applied to detect and correct errors [9, 35, 47, 48].

Portuguese is the 9th most spoken language in the world [3] and has a rich history and cultural heritage, with millions of historical documents stored across various archives and libraries. Since the early 2000s, efforts have been made to digitise these records, and currently, over 70 million documents are available¹. However, most of these digital documents remain in image format, making them difficult to search. Moreover, many of these documents already show signs of degradation, meaning that simple OCR-based text extraction may not be sufficient.

Despite being a well-researched field, post-OCR processing systems primarily focus on English, leaving other languages under-represented. For the Portuguese language, research on post-OCR processing is scarce, and there is a significant lack of resources for training and evaluation. To address this, we introduce PORTO, a publicly^{2,3} containing documents from the 17th to the 20th centuries, totalling 3,782 pages and 886.4 thousand tokens, with transcriptions systematically reviewed and corrected.

2 Related Work

The development and evaluation of post-OCR processing systems depend significantly on datasets that pair OCR outputs or images with their corresponding transcriptions. Several datasets have been compiled for European languages, ranging from monolingual to multilingual collections.



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¹<https://dglab.gov.pt/area-arquivos/>

²<https://huggingface.co/datasets/LIACC/PORTO>

³<https://doi.org/10.5281/zenodo.16889416>

Monolingual examples include, for Dutch, DBNL [11] (220 books, 18th-19th) and OCR-GT [49] (2 thousand pages, 18th-20th). For English, OCR-IDL [6] (26 million pages, 20th), RDD [14] (2.2 million lines, 16th-19th), TCP [14] (8.6 million lines, 19th), Overproof-2 [17] (49 thousand words, 19th-20th), Overproof-3 [17] (18 thousand words, 19th-20th), Hegghammer [26] (322 pages, 19th-20th), TREC-5 [27] (55.6 thousand documents, 20th), MiBio [30, 31] (211 documents, 20th), ALTA 2017 [32] (8 thousand documents, 19th-21th), ACL collection [33] (19 thousand documents, 20th-21th), IMPACT [39] (388 pages, 15th-20th). For French OCR17 [20] (37 documents, 17th). For Greek Polyton-DB [45] (38 pages, 19th-20th) and GRPOLY-DB [22] (399 pages, 19th-20th).

Multi-lingual datasets are also available, such as ICDAR 2017 [8] (6 million characters), ICDAR 2019 [42] (200 pages), ENP [10] (528 pages), CAMIO [4] (1,250 pages), Text+Berg [24] (16 thousand articles, 19th-21th), RETAS [50] (160 books), GT4HistOCR [46] (110 books, 15th-19th).

These datasets vary significantly in size, ranging from just a few sentences to several thousand pages. For Portuguese, we have identified three datasets: REGIS [12], a dataset containing approximately 21 thousand geoscientific documents in Portuguese, stored in PDF format and produced between 1957 and 2020. Despite its extensive size, it provides ground truth annotations for only 170 sentences, each averaging around 30 tokens. EPISA [18], a dataset comprising 708 images with manually transcribed typewritten documents, including letters, structured and unstructured reports, process covers, and theatre play covers from the 20th century. The EPISA contains a total of 69,000 tokens [38]. Finally, ESTER-Pt [44] consists of three versions. The first is a synthetic text-based dataset created from Wikipedia-Pt, with artificially inserted errors that mimic typical OCR patterns. The second is a synthetic image-based dataset, using the same Wikipedia texts to generate images that replicate the appearance of scanned pages. The third version contains real image-based PDFs of scanned books written between 1800 and 1910. As observed, the existing datasets exhibit low variability, particularly in terms of their historical period.

3 Corpus

The PORTO dataset comprises scanned documents and their corresponding transcriptions, collected from four sources: the Tycho Brahe Corpus [21], *Corpus Histórico da Linguagem da Medicina em Português* (CHLMP) [2, 19], *Arquivo dos Açores* (AdA) [1], and ESTER-Pt [44].

The ESTER-Pt real image-based version contains nine books – *Annos de Prosa* (AdP), *Carlota Angela* (CA), *Cidade do Vicio* (CdV), *Historias Brasileiras* (HB), *Historias sem Data* (HSD), *Nossa Gente* (NG), *O Guarany* (OG), *Poesias Herculanio* (PH), and *Reliquias de Casa Velha* (RdCV) –, primarily novels and poetry, written between 1800 and 1910. The scanned versions were obtained from Acervo Digital⁴, while the transcriptions came from Project Gutenberg⁵.

The Tycho Brahe Corpus [21] includes 95 Portuguese texts from over 50 authors born between the 14th and 19th centuries. From this collection, we have access to two scanned books and their transcriptions: *Regras da Lingua Portuguesa*, *Espelho da Lingua*

Table 1: PORTO dataset overview.

Corpus	Book	Date	Pages	Tokens
CHLMP	OMD	1707	569	185.2k
	PRAE	1741	101	16.6k
ESTER-Pt	AdP	1863	261	67.5k
	CA	1858	160	49.7k
	CdV	1882	312	69.5k
	HB	1874	212	45.6k
	HSD	1884	272	52.6k
	NG	1900	218	44.5k
	OG	1857	49	8.6k
	PH	1860	308	33.8k
	RdCV	1906	236	49.3k
AdA	Vol. 1	1878	499	185.1k
Tycho	RLPELL	1676	378	48.0k
	IldC	1929	207	30.4k
Total		17th-20th	3,782	886.4k

Latina (RLPELL), written by Jerónimo Contador de Argote in 1676, and *Iracema, lenda do Ceará* (IldC), written by José de Alencar in 1929.

CHLMP contains transcriptions of two historical medical books: *Observações medicas doutrinaes de cem casos gravissimos* (OMD), written by João Curvo Semedo in 1707, which consists of 101 chapters spanning 634 pages, and *Postilla Religiosa, e Arte de Enfermeiros* (PRAE), authored by Diogo de Santiago in 1741, containing 59 chapters across 340 pages. The first book documents 101 medical cases, providing valuable insights into 18th-century healthcare and medical practices, while the second explores the intersection of religion and infirmery work, with a particular focus on medical procedures and end-of-life care.

AdA is a reference for historical research on the Azores archipelago, comprising two series with a total of 20 volumes. The first series, published between 1878 and 1959, contains 15 volumes, 12 of which have been digitised by the Internet Archive⁶. Due to numerous OCR errors in the digitised texts, the University of Azores curated the first volume. This volume spans 594 pages and covers a significant historical period from 1439 to 1877, featuring correspondence and documents from these centuries.

The ESTER-Pt dataset makes available image-transcription page pairs, while the other sources provide for each book the transcriptions consolidated into a single text file and the corresponding scanned images compiled into a single PDF.

Table 1 presents a summary of the PORTO dataset by book, including the number of pages, token counts, and the publication year. We now proceed to detail the process of creating the dataset.

3.1 Methodology

We aimed to provide image-transcription pairs for training and evaluating OCR and post-OCR correction systems. Our goal was to keep the transcriptions as faithful as possible to the originals, preserving any orthographic errors without correction attempts.

⁴<https://bndigital.bn.gov.br/acervodigital/>

⁵<https://www.gutenberg.org/>

⁶archive.org

Similar to ESTER-Pt, the PORTO dataset includes one image-transcription pair per page of each book. During this process, any textual elements present in the images that are not contained in the transcriptions – such as notes, footnotes, and page numbers/letters – were manually struck out by applying black redaction marks. After establishing the image-transcription pairs, we conducted a quality assessment to identify and correct any transcription errors, given the uncertainty of the initial transcription quality.

The PORTO dataset was initially developed using material exclusively from the CHLMP collection, which allowed us to perform a full manual review of those books. As we expanded the dataset to include additional sources, reviewing every page was not feasible given the significant time and effort required. So, we adopted a stratified sampling approach, grouping the pages of each book into quartiles based on Word Error Rate (WER) between the OCR text and the corresponding transcriptions. WER is a quality metric that measures the minimum number of operations (insertions, deletions, and substitutions) required to convert, for example, an OCR'ed text into the transcription. WER is computed as follows:

$$\text{Word Error Rate} = \frac{S + D + I}{N} * 100 \quad (1)$$

where S , D and I are the number of substitutions, deletions, and insertions, respectively, and N is the number of words in the transcription. For each book, we evaluated multiple OCR engines and selected the one with the lowest average WER to generate the quartiles. During the calculation of WER, the texts are normalised by lowercasing, removing punctuation and diacritics, standardising white-spaces, and applying character substitutions. Specifically, 'ç' is replaced with 'c', 'j' with 's', 'Æ' and 'æ' with 'ae', '&' with 'e', 'o' with 'o', and 'a' with 'a'.

From each quartile, we randomly sampled at least 10% of the pages, ensuring a minimum of ten pages per quartile. The sampled transcriptions were manually reviewed, and the WER was then calculated by comparing the source transcriptions with the manually corrected versions. If the average WER between both exceeded 1.5% in a given quartile, we manually reviewed all transcriptions within that quartile. This methodology focused the manual effort on the most error-prone quartiles.

To ensure comprehensive quality assessment, we assessed all quartiles, as transcription quality may vary throughout a book, and errors are not necessarily concentrated in just the worst-performing quartiles. By adopting this stratified sampling approach, we ensured a more adaptive and thorough quality control process.

3.2 Optical Character Recognition

In this work, we applied the following OCR engines: Tesseract⁷, Azure AI Document Intelligence (AADI)⁸, Transkribus (TKBUS)⁹, and PaddleOCR¹⁰. AADI is a proprietary OCR service within the Azure ecosystem, designed to extract printed and handwritten text from PDF documents and scanned images. Tesseract is an open-source OCR engine initially developed by Hewlett-Packard between 1984 and 1994 and later released as open-source in 2005, with Google

⁷<https://github.com/tesseract-ocr/tesseract>

⁸<https://azure.microsoft.com/en-us/products/ai-services/ai-document-intelligence>

⁹<https://www.transkribus.org/>

¹⁰<https://github.com/Mushroomcat9998/PaddleOCR>

Table 2: WER comparing source transcriptions with OCR outputs.

Corpus	Book	AADI	PaddleOCR	Tesseract	TKBUS
CHLMP	OMD	24.09	41.96	39.31	12.75
	PRAE	16.96	25.64	23.73	7.42
ESTER-Pt	AdP	3.19	11.48	7.54	6.09
	CA	0.98	2.04	3.26	3.22
	CdV	1.10	3.11	2.06	1.42
	HB	1.63	5.52	6.73	6.52
	HSD	2.15	6.14	11.97	5.44
	NG	1.57	6.26	3.89	2.82
	OG	10.91	23.72	19.30	31.99
	PH	2.82	5.78	5.22	6.18
	RdCV	2.00	3.81	16.34	3.53
AdA	Vol. 1	3.05	13.80	8.14	6.68
Tycho	RLPELL	27.70	42.92	40.39	19.15
	IldC	5.33	10.04	9.13	6.73
Avg.		8.72	16.92	16.14	8.01

sponsorship since 2006. TKBUS is specialised in historical documents. It was developed through two EU-funded research projects and is now maintained by READ-COOP SCE. In our experiments, we used the extended multi-language print model M1, which includes support for antiqua, blackletter, typewriters, computer print-outs, and decorative fonts. PaddleOCR is an OCR toolkit based on PaddlePaddle, supporting 80 languages, including Portuguese.

3.3 Transcription Correction

Table 2 presents a comparative analysis of OCR performance across engines. Our evaluation shows that AADI achieves the lowest WER in ESTER-Pt, AdA, and the IldC book from Tycho, while TKBUS outperforms on the remaining books. Notably, AADI tends to perform better on more recently published documents, while TKBUS yields superior results on older ones. We subsequently used these best-performing engines to generate the quartiles for each book as specified in our methodology.

During manual inspection of the pages with the highest WER – relative to the best-performing OCR engine – we observed that some contain complex layouts, such as double-column text mixed within single-column text (see Figure 1). Since there is no standardised convention for representing such layouts and the variation in word ordering can significantly impact the WER metric [34], we excluded from transcription correction any pages exhibiting these complex layouts, while still retaining them within the PORTO dataset. This exclusion applies to 3 pages from ESTER-Pt, 77 from Tycho, and 7 from AdA. Details regarding these complex layout pages are provided in the metadata accompanying the PORTO dataset.

The remaining image-transcription pairs for each book were grouped into quartiles based on WER. We then proceeded with the correction process, focusing primarily on misidentified letters and numbers while also addressing punctuation and diacritical marks when discrepancies were identified. It is important to note that punctuation and diacritics were sometimes difficult to verify due to

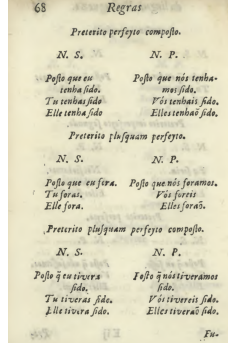


Figure 1: Complex layout example from RLPELL book.

Table 3: WER between original and reviewed transcriptions by quartile, with corresponding review coverage percentages. Q denotes quartile number; R indicates the percentage of transcriptions reviewed within each quartile.

Corpus	Book	Q1 WER/R	Q2 WER/R	Q3 WER/R	Q4 WER/R
CHLMP	OMD	0.8 / 100	1.3 / 100	1.6 / 100	3.3 / 100
	PRAE	0.0 / 100	0.0 / 100	0.0 / 100	0.1 / 100
ESTER-Pt	AdP	0.0 / 18	0.2 / 38	0.3 / 39	23.5 / 100
	CA	0.0 / 30	0.0 / 28	0.1 / 30	0.0 / 25
	CdV	0.0 / 14	0.0 / 13	0.1 / 13	0.8 / 13
	HB	0.1 / 19	0.2 / 23	0.4 / 26	0.4 / 19
	HSD	0.2 / 15	0.5 / 16	0.9 / 15	1.4 / 15
	NG	0.0 / 18	0.0 / 19	0.2 / 19	0.6 / 20
	OG	2.7 / 100	3.6 / 100	6.0 / 100	15.5 / 100
	PH	0.0 / 33	0.5 / 100	0.1 / 29	3.1 / 100
	RdCV	0.0 / 16	0.0 / 21	0.3 / 24	0.5 / 16
AdA	Vol. 1	0.2 / 15	0.6 / 22	1.3 / 39	2.8 / 100
Tycho	RLPELL	0.7 / 49	0.8 / 72	1.2 / 67	1.8 / 100
	IldC	1.1 / 100	2.5 / 100	3.6 / 100	369.7 / 100

image damage. In cases where we were uncertain about the correct form, we chose to retain the version from the source transcription.

Table 3 presents the WER between the sourced transcription and the reviewed version, as well as the percentage of transcriptions reviewed within each quartile. The analysis revealed that 13 quartiles contained transcription issues with an average WER greater than 1.5%; of these, 3 exhibited severe errors with an average WER exceeding 15%. To ensure data quality, all problematic quartiles were subject to full review.

In total, we reviewed 2,017 pages, representing 53.3% of the entire dataset. Across these pages, we corrected approximately 59.2 thousand tokens – comprising insertions, substitutions, and deletions – which represents 6.6% of the total tokens.

During our review, we identified several pages written in languages other than Portuguese. To detect these variations, we employed `langdetect`¹¹, a Python package that uses a naive Bayesian filter with over 99% precision for 53 languages, Portuguese included.

¹¹<https://pypi.org/project/langdetect/>

Table 4: WER comparing PORTO dataset with OCR outputs.

Corpus	Book	AADI	PaddleOCR	Tesseract	TKBUS
CHLMP	OMD	22.94	41.13	38.37	11.22
	PRAE	16.96	25.64	23.72	7.40
ESTER-Pt	AdP	2.67	10.99	7.04	5.58
	CA	0.97	2.03	3.25	3.22
	CdV	1.07	3.09	2.04	1.41
	HB	1.58	5.47	6.67	6.48
	HSD	2.07	6.04	11.91	5.37
	NG	1.55	6.23	3.86	2.80
	OG	3.43	17.17	12.41	25.53
	PH	1.79	4.57	3.98	4.92
	RdCV	1.97	3.78	16.33	3.52
AdA	Vol. 1	2.29	12.37	7.72	5.74
Tycho	RLPELL	21.79	40.61	37.53	13.66
	IldC	1.63	6.98	6.56	3.11
Avg.		7.14	15.48	14.84	6.53

Our analysis revealed that RLPELL contains one page in Latin; RdCV includes one page with a mixture of French and Portuguese; and AdA comprises one page in Latin, four in French, and twelve in Spanish. All language identification details are documented in the metadata accompanying the PORTO dataset. Additionally, it is important to note that the Tycho dataset lacks punctuation in the majority of its transcriptions.

Finally, Table 4 summarises OCR performance on the PORTO dataset, demonstrating a consistent reduction in WER across all OCR engines compared to the initial evaluation (see Table 2). As expected, improvements are particularly pronounced in books identified with problematic quartiles during our review process.

4 Conclusion

The PORTO dataset significantly increases the quantity of transcribed pages, adding 1,754 new ones and correcting erroneous transcriptions from ESTER-Pt. It also expands the available periods by introducing 5 new books spanning from the 17th to the 20th century. Alongside the transcriptions, we also provide OCR outputs generated by two open-source and two proprietary engines. These outputs offer a valuable resource for research into OCR error patterns and the development of post-OCR correction methods.

As future work, this dataset presents valuable opportunities for developing post-OCR correction systems tailored to historical Portuguese. Another promising direction is to extend multi-lingual post-OCR correction systems to support historical Portuguese.

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GenAI Usage Disclosure

This paper and the PORTO dataset were developed without use of Generative AI.

References

- [1] [n. d.]. Arquivo dos Açores. <https://hdl.handle.net/21.11129/0000-000D-F8C0-2>. Accessed: 16-5-2023.
- [2] [n. d.]. Corpus Histórico da Linguagem da Medicina em Português (Século XVIII): Terminologia Diacrônica e Humanidades Digitais. <https://sites.google.com/view/projeto38597>. Accessed: 16-5-2023.
- [3] [n. d.]. What are the top 200 most spoken languages? <https://www.ethnologue.com/guides/ethnologue200>. Accessed: 26-01-2023.
- [4] Michael Arrigo, Stephanie Strassel, Nolan King, Thao Tran, and Lisa Mason. 2022. CAMIO: A Corpus for OCR in Multiple Languages. In *Proceedings of the Thirteenth Language Resources and Evaluation Conference*, Nicoletta Calzolari, Frédéric Bèchet, Philippe Blache, Khalid Choukri, Christopher Cieri, Thierry Declercq, Sara Goggi, Hitoshi Isahara, Bente Maegaard, Joseph Mariani, Hélène Mazon, Jan Odijk, and Stelios Piperidis (Eds.). European Language Resources Association, Marseille, France, 1209–1216. <https://aclanthology.org/2022.lrec-1.129/>
- [5] Guilherme Torresan Bazzo, Gustavo Acauan Lorentz, Danny Suarez Vargas, and Viviane P. Moreira. 2020. Assessing the Impact of OCR Errors in Information Retrieval. In *Advances in Information Retrieval*, Joemon M. Jose, Emine Yilmaz, João Magalhães, Pablo Castells, Nicola Ferro, Mário J. Silva, and Flávio Martins (Eds.). Springer International Publishing, Cham, 102–109.
- [6] Ali Furkan Bilen, Rubèn Tito, Lluís Gomez, Ernest Valveny, and Dimosthenis Karatzas. 2023. OCR-IDL: OCR Annotations for Industry Document Library Dataset. In *Computer Vision – ECCV 2022 Workshops*, Leonid Karlinsky, Tomer Michaeli, and Ko Nishino (Eds.). Springer Nature Switzerland, Cham, 241–252.
- [7] Syed Saqib Bukhari, Ahmad Kadi, Mohammad Ayman Jouneh, Fahim Mahmood Mir, and Andreas Dengel. 2017. anyOCR: An Open-Source OCR System for Historical Archives. *14th IAPR International Conference on Document Analysis and Recognition (ICDAR)* 1 (11 2017), 305–310. doi:10.1109/ICDAR.2017.58
- [8] Guillaume Chiron, Antoine Doucet, Mickael Coustaty, and Jean-Philippe Moreux. 2017. ICDAR 2017 Competition on Post-OCR Text Correction. *2017 14th IAPR International Conference on Document Analysis and Recognition (ICDAR)* 1 (11 2017), 1423–1428. doi:10.1109/ICDAR.2017.232
- [9] Guillaume Chiron, Antoine Doucet, Mickael Coustaty, Muriel Visani, and Jean-Philippe Moreux. 2017. Impact of OCR Errors on the Use of Digital Libraries: Towards a Better Access to Information. *ACM/IEEE Joint Conference on Digital Libraries (JCDL)* (6 2017), 1–4. doi:10.1109/JCDL.2017.7991582
- [10] Christian Clausner, Christos Papadopoulos, Stefan Pletschacher, and Apostolos Antonacopoulos. 2015. The ENP image and ground truth dataset of historical newspapers. In *2015 13th International Conference on Document Analysis and Recognition (ICDAR)*, 931–935. doi:10.1109/ICDAR.2015.7333898
- [11] DBNL. 2019. DBNL OCR Data set. doi:10.5281/zenodo.3239290
- [12] Lucas Lima de Oliveira, Danny Suarez Vargas, Antônio Marcelo Azevedo Alexandre, Fábio Corrêa Cordeiro, Diogo da Silva Magalhães Gomes, Max de Castro Rodrigues, Regis Krueel Romeu, and Viviane Pereira Moreira. 2023. Evaluating and mitigating the impact of OCR errors on information retrieval. *International Journal on Digital Libraries* 24, 1 (March 2023), 45–62. doi:10.1007/s00799-023-00345-6
- [13] Eva D'hondt, Cyril Grouin, and Brigitte Grau. 2017. Generating a Training Corpus for OCR Post-Correction Using Encoder-Decoder Model. *Proceedings of the Eighth International Joint Conference on Natural Language Processing (Volume 1: Long Papers)* (2017), 1006–1014.
- [14] Rui Dong and David Smith. 2018. Multi-Input Attention for Unsupervised OCR Correction. *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (2018), 2363–2372. doi:10.18653/v1/P18-1220
- [15] Haimonti Dutta and Aayushee Gupta. 2022. PNRank: Unsupervised ranking of person name entities from noisy OCR text. *Decision Support Systems* 152 (2022), 113662. doi:10.1016/j.dss.2021.113662
- [16] Maud Ehrmann, Ahmed Hamdi, Elvys Linhares Pontes, Matteo Romanello, and Antoine Doucet. 2023. Named Entity Recognition and Classification in Historical Documents: A Survey. *ACM Comput. Surv.* 56, 2, Article 27 (Sept. 2023), 47 pages. doi:10.1145/3604931
- [17] John Evershed and Kent Fitch. 2014. Correcting noisy OCR: Context beats Confusion. *Proceedings of the First International Conference on Digital Access to Textual Cultural Heritage - DATeCH '14* (2014), 45–51. doi:10.1145/2595188.2595200
- [18] Margarida Falcão. 2022. Manual Transcriptions of Typewritten Digital Representations of Portuguese Cultural Heritage Documents from the 20th Century. <https://rdm.inescet.pt/dataset/cs-2022-005>. doi:10.25747/WPNA-JE39
- [19] Maria José Finatto, Paulo Quaresma, and Maria Filomena Gonçalves. 2018. Portuguese corpora of the 18th century: old Medicine texts for teaching and research. *Proceedings of the Conference on Language Technologies & Digital Humanities* (2018).
- [20] Simon Gabay, Thibault Clérice, and Christian Reul. 2023. OCR17: Ground Truth and Models for 17th c. French Prints (and hopefully more). *Journal of Data Mining & Digital Humanities* 2023, Dataset (June 2023), 6492. doi:10.46298/jdmhdh.6492
- [21] Charlotte Galves. 2018. The Tycho Brahe Corpus of Historical Portuguese: Methodology and results. *Linguistic Variation* 18 (07 2018), 49–73. doi:10.1075/lv.00004.gal
- [22] Basilis Gatos, Nikolaos Stamatopoulos, Georgios Louloudis, Giorgos Sfikas, George Retsinas, Vassilis Papavassiliou, Fotini Sunistira, and Vassilis Katsouras. 2015. GRPOLY-DB: An old Greek polytonic document image database. In *2015 13th International Conference on Document Analysis and Recognition (ICDAR)*, 646–650. doi:10.1109/ICDAR.2015.7333841
- [23] Amit Gupta, Alexey Romanov, Sahitya Mantravadi, Dalitso Banda, Jianjie Liu, Raza Khan, Lakshmanan Ramu Meenal, Benjamin Han, and Soundar Srinivasan. 2021. Lights, Camera, Action! A Framework to Improve NLP Accuracy over OCR documents. *ArXiv abs/2108.02899* (2021). <https://api.semanticscholar.org/CorpusID:236950757>
- [24] Anne Göhring and Martin Volk. 2011. The Text+Berg Corpus An Alpine French-German Parallel Resource. *Actes de la 18e conférence sur le Traitement Automatique des Langues Naturelles. Articles courts* (2011), 97–102. <http://www.textberg.ch/index.php?id=4&lang=en>
- [25] Ahmed Hamdi, Axel Jean-Caurant, Nicolas Sidère, Mickaël Coustaty, and Antoine Doucet. 2020. Assessing and Minimizing the Impact of OCR Quality on Named Entity Recognition. In *Digital Libraries for Open Knowledge*, Mark Hall, Tanja Merčun, Thomas Risse, and Fabien Duchateau (Eds.). Springer International Publishing, Cham, 87–101.
- [26] Thomas Hegghammer. 2022. OCR with Tesseract, Amazon Textract, and Google Document AI: a benchmarking experiment. *Journal of Computational Social Science* 5, 1 (May 2022), 861–882. doi:10.1007/s42001-021-00149-1
- [27] Paul B. Kantor and Ellen M. Voorhees. 2000. The TREC-5 confusion track: Comparing retrieval methods for scanned text. *Information Retrieval* 2 (2000), 165–176. doi:10.1023/a:1009902609570
- [28] Kimmo Kettunen, Heikki Keskustalo, Sanna Kumpulainen, Tuula Pääkkönen, and Juha Rautiainen. 2022. OCR quality affects perceived usefulness of historical newspaper clippings – a user study. *arXiv:2203.03557 [cs.IR]* <https://arxiv.org/abs/2203.03557>
- [29] Ido Kissos and Nachum Dershowitz. 2016. OCR Error Correction Using Character Correction and Feature-Based Word Classification. *2th IAPR Workshop on Document Analysis Systems (DAS)* (4 2016), 198–203. doi:10.1109/DAS.2016.44
- [30] Jie Mei, Aminul Islam, Abidrahman Moh'd, Yajing Wu, and Evangelos Milios. 2017. Post-Processing OCR Text using Web-Scale Corpora. *DocEng '17: Proceedings of the 2017 ACM Symposium on Document Engineering* (8 2017), 117–120. doi:10.1145/3103010.3121032
- [31] Jie Mei, Aminul Islam, Abidrahman Moh'd, Yajing Wu, and Evangelos Milios. 2018. Statistical learning for OCR error correction. *Information Processing & Management* 54 (11 2018), 874–887. Issue 6. doi:10.1016/j.ipm.2018.06.001
- [32] Diego Mollá and Steve Cassidy. 2017. Overview of the 2017 ALTA Shared Task: Correcting OCR Errors. *Proceedings of the Australasian Language Technology Association Workshop 2017* (2017), 115–118. <https://www.nla.gov.au/>
- [33] Vivi Nastase and Julian Hitschler. 2018. Correction of OCR Word Segmentation Errors in Articles from the ACL Collection through Neural Machine Translation Methods. *Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018)* (2018). <http://www.cl.uni-heidelberg.de/english/>
- [34] Clemens Neudecker, Konstantin Baierer, Mike Gerber, Christian Clausner, Apostolos Antonacopoulos, and Stefan Pletschacher. 2021. A survey of OCR evaluation tools and metrics. In *Proceedings of the 6th International Workshop on Historical Document Imaging and Processing* (Lausanne, Switzerland) (HIP '21). Association for Computing Machinery, New York, NY, USA, 13–18. doi:10.1145/3476887.3476888
- [35] Thi-Tuyet-Hai Nguyen, Mickael Coustaty, Antoine Doucet, Adam Jatowt, and Nhu-Van Nguyen. 2018. Adaptive Edit-Distance and Regression Approach for Post-OCR Text Correction. *Maturity and Innovation in Digital Libraries* (2018), 278–289. doi:10.1007/978-3-030-04257-8_29
- [36] Thi Tuyet Hai Nguyen, Adam Jatowt, Mickael Coustaty, and Antoine Doucet. 2022. Survey of Post-OCR Processing Approaches. *Comput. Surveys* 54 (7 2022), 1–37. Issue 6. doi:10.1145/3453476
- [37] Brian Ogilvie. 2016. Scientific Archives in the Age of Digitization. *ISIS* 107 (3 2016), 77–85. Issue 1. doi:10.1086/686075
- [38] Tomás Freitas Osório and Henrique Lopes Cardoso. 2024. Historical Portuguese corpora: a survey. *Language Resources and Evaluation* (July 2024). doi:10.1007/s10579-024-09757-5
- [39] Christos Papadopoulos, Stefan Pletschacher, Christian Clausner, and Apostolos Antonacopoulos. 2013. The IMPACT dataset of historical document images. *Proceedings of the 2nd International Workshop on Historical Document Imaging and Processing (HIP '13)* (2013), 123. doi:10.1145/2501115.2501130

- [40] James Philips and Nasseh Tabrizi. 2020. Historical Document Processing: A Survey of Techniques, Tools, and Trends. *Proceedings of the 12th International Joint Conference on Knowledge Discovery, Knowledge Engineering and Knowledge Management* 1 (2020), 341–349. doi:10.5220/0010177403410349
- [41] Roswitha Poll. 2010. NUMERIC: statistics for the digitisation of European cultural heritage. *Program: electronic library and information systems* 44 (4 2010), 122–131. Issue 2. doi:10.1108/00330331011039481
- [42] Christophe Rigaud, Antoine Doucet, Mickael Coustaty, and Jean-Philippe Moreux. 2019. ICDAR 2019 Competition on Post-OCR Text Correction. *International Conference on Document Analysis and Recognition (ICDAR)* (9 2019), 1588–1593. doi:10.1109/ICDAR.2019.00255
- [43] Shruti Rijhwani, Antonios Anastasopoulos, and Graham Neubig. 2020. OCR Post Correction for Endangered Language Texts. *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (2020), 5931–5942. doi:10.18653/v1/2020.emnlp-main.478
- [44] Moniele Kunrath Santos, Guilherme Bazzo, Lucas Lima de Oliveira, and Viviane Pereira Moreira. 2023. ESTER-Pt: An Evaluation Suite for TExt Recognition in Portuguese. In *Document Analysis and Recognition - ICDAR 2023*, Gernot A. Fink, Rajiv Jain, Koichi Kise, and Richard Zanibbi (Eds.). Springer Nature Switzerland, Cham, 366–383.
- [45] Fotini Simistira, Adnan Ul-Hassan, Vassilis Papavassiliou, Basilis Gatos, Vassilis Katsouras, and Marcus Liwicki. 2015. Recognition of historical Greek polytonic scripts using LSTM networks. In *2015 13th International Conference on Document Analysis and Recognition (ICDAR)*. 766–770. doi:10.1109/ICDAR.2015.7333865
- [46] Uwe Springmann, Christian Reul, Stefanie Dipper, Johannes Baiter Bayerische, and Staatsbibliothek München. 2018. Ground Truth for training OCR engines on historical documents in German Fraktur and Early Modern Latin. *Journal for Language Technology and Computational Linguistics (JLCL)* 33 (2018), 97–114. Issue 1. <http://www.cis.uni-muenchen.de/ocrworkshop/program.html>
- [47] Myriam C. Traub, Jacco van Ossenbruggen, and Lynda Hardman. 2015. Impact Analysis of OCR Quality on Research Tasks in Digital Archives. *Research and Advanced Technology for Digital Libraries. TPDL 2015. Lecture Notes in Computer Science* 9316 (2015), 252–263. doi:10.1007/978-3-319-24592-8_19
- [48] Daniel Van Strien, Kaspar Beelen, Mariona Ardanuy, Kasra Hosseini, Barbara McGillivray, and Giovanni Colavizza. 2020. Assessing the Impact of OCR Quality on Downstream NLP Tasks. In *Proceedings of the 12th International Conference on Agents and Artificial Intelligence*. SCITEPRESS - Science and Technology Publications, Valletta, Malta, 484–496. doi:10.5220/0009169004840496
- [49] L. Wilms, R. Nijssen, and T. Koster. 2020. Historical newspaper OCR ground-truth data set. KB Lab: The Hague.
- [50] Ismet Zeki Yalniz and R. Manmatha. 2011. A Fast Alignment Scheme for Automatic OCR Evaluation of Books. *International Conference on Document Analysis and Recognition* (9 2011), 754–758. doi:10.1109/ICDAR.2011.157
- [51] Elaine Zosa, Stephen Mutuvi, Mark Granroth-Wilding, and Antoine Doucet. 2021. Evaluating the Robustness of Embedding-Based Topic Models to OCR Noise. In *Towards Open and Trustworthy Digital Societies*, Hao-Ren Ke, Chei Sian Lee, and Kazunari Sugiyama (Eds.). Springer International Publishing, Cham, 392–400.